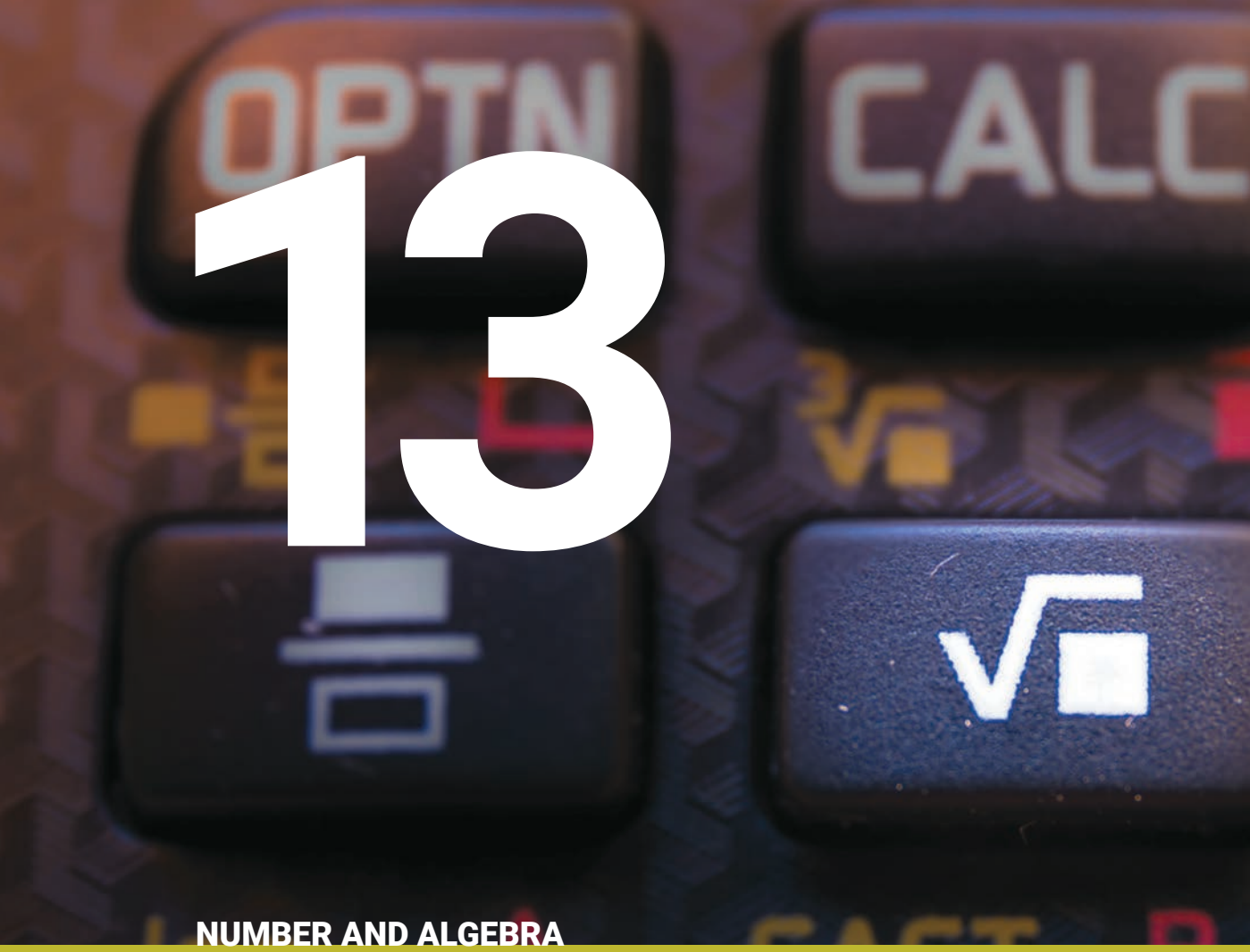


13

NUMBER AND ALGEBRA

SURDS

OPTIONAL PATH TOPIC
RECOMMENDED FOR STAGE 6 MATHEMATICS ADVANCED

When applying Pythagoras' theorem, we have found lengths that cannot be expressed as an exact rational number. Pythagoras encountered this when calculating the diagonal of a square of side length 1 unit. A **surd** is a square root ($\sqrt{\ }$), cube root ($\sqrt[3]{\ }$), or any type of root whose exact decimal or fraction value cannot be found.

The Pythagoreans explained nature, the universe — in fact everything — in terms of whole numbers. Apparently, they were so upset about the discovery of surds that they tried to keep the discovery a secret. Hippasus, one of the Pythagoreans, was drowned for revealing the secret to outsiders.



iStock.com/Lamberto Jesus

Chapter outline

	Working mathematically				
13.01 Surds*	U	F		R	C
13.02 Simplifying surds*	U	F		R	
13.03 Adding and subtracting surds*	U	F			
13.04 Multiplying and dividing surds*	U	F			
13.05 Binomial products involving surds*	U	F			
13.06 Rationalising the denominator*	U	F			

*PATH

Note: Chapter 12, Products and factors, must be completed before this chapter.

Wordbank

irrational number A number such as π or $\sqrt{2}$ that cannot be expressed as a fraction $\frac{a}{b}$

rational number Any number that can be written in the form $\frac{a}{b}$, where a and b are integers and $b \neq 0$

rationalise the denominator To simplify a fraction involving a surd by making its denominator rational (that is, not a surd)

real number A number that is either rational or irrational and whose value can be graphed on a number line

simplify a surd To write a surd $\sqrt{x}$ in its simplest form so that x has no factors that are perfect squares

surd A square root (or other root) whose exact value cannot be found

Quiz
Wordbank 13

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Videos (4):
13.02 Simplifying surds 2
• Simplifying surds 1
13.05 Binomial products involving surds
13.06 Rationalising the denominator

Quizzes (2):
• Wordbank 13
• SkillCheck 13

Worksheets (2):
13.01 Surds on the number line
13.04 Multiplying and dividing surds

Puzzles (3):
13.02 Simplifying surds
13.03 Surds code puzzle
13.04 Surds

Technology (1):
13.02 Simplifying surds quiz

Spreadsheet (1):
13.02 Simplifying surds quiz

Nelson MindTap

To access resources above, visit
cengage.com.au/nelsonmindtap

In this chapter you will:

- (PATH) describe real, rational and irrational numbers and surds
- (PATH) simplify, add, subtract, multiply and divide surds
- (PATH) expand and simplify binomial products involving surds
- (PATH) rationalise the denominator of expressions of the form $\frac{a\sqrt{b}}{c\sqrt{d}}$

PATH

Quiz
SkillCheck 13

SkillCheck ANSWERS ON P. 637

1 Simplify each expression.

a $(5y)^2$ **b** $(4m)^3$ **c** $(-3x)^2$

2 Expand each expression.

a $5(x + 2)$ **b** $4(y - 3)$ **c** $3(1 + 2w)$
d $2(5 - y)$ **e** $-5(2a + 3)$ **f** $k(1 + 2k)$

3 Select the square numbers from the following list of numbers.

44 81 25 100 75
72 16 50 64 32

4 Expand and simplify each expression.

a $(m + 3)(m + 7)$ **b** $(y + 1)(y - 4)$ **c** $(n - 2)(n - 3)$
d $(2d + 3)(1 + 3d)$ **e** $(1 - 5p)(4 + 3p)$ **f** $(3a + 2f)(a + 5f)$
g $(x + 4)^2$ **h** $(y - 3)^2$ **i** $(2k + 1)^2$
j $(a - 5)(a + 5)$ **k** $(t + 7)(t - 7)$ **l** $(3m + 4)(3m - 4)$

Surds

13.01

PATH

13.01

A **surd** is a square root ($\sqrt{\ }$), cube root ($\sqrt[3]{\ }$), or any type of root whose exact decimal or fraction value cannot be found. As a decimal, its digits run endlessly *without repeating* (like π), so they are neither terminating nor recurring decimals.

$\sqrt{7}$ is read as 'the square root of 7' or simply 'root 7'.

$\sqrt{7}$ is called the **exact value**, $\sqrt{7} \approx 2.64575$ is the **approximate value**.

Rational numbers such as fractions, decimals and percentages, can be expressed in the form $\frac{a}{b}$, where a and b are integers ($b \neq 0$), but surds are **irrational numbers** because they cannot be expressed in this form.

Rational numbers

can be expressed in the form $\frac{a}{b}$

Integers

$\frac{4}{1} = 4, \frac{26}{1} = 26, \frac{-3}{1} = -3$

Terminating decimals

0.5, $7\frac{1}{8} = 7.125$,

16% = 0.16, 1.32

Recurring decimals

$\frac{2}{3} = 0.666\ldots$

$\frac{5}{6} = 0.833\ldots$

$\frac{4}{11} = 0.3636\ldots$

Irrational numbers

cannot be expressed in the form $\frac{a}{b}$

Surds

$\sqrt{5}, -\sqrt{2}, \frac{\sqrt{11}}{3}, 8\sqrt{6}$

Transcendental numbers

Have no pattern and are non-recurring

e.g. $\pi = 3.14159...$,

$\cos 38° = 0.78801...$, $e = 2.71828...$

Example 1

Select the surds from this list of square roots:

$\sqrt{72}$ $\sqrt{121}$ $\sqrt{64}$ $\sqrt{90}$ $\sqrt{28}$

Solution

$\sqrt{72} = 8.4852...$ $\sqrt{121} = 11$ $\sqrt{64} = 8$

$\sqrt{90} = 9.4868...$ $\sqrt{28} = 5.2915...$

So the surds are $\sqrt{72}$, $\sqrt{90}$ and $\sqrt{28}$.

PATH

Example 2

Is each number rational or irrational?

a $4\frac{2}{5}$ **b** $\sqrt[3]{-8}$ **c** $-\sqrt{7}$ **d** $0.\dot{6}$ **e** 5π

Solution

a $4\frac{2}{5} = \frac{22}{5}$ which is in the form of a fraction $\frac{a}{b}$

$\therefore 4\frac{2}{5}$ is a rational number.

b $\sqrt[3]{-8} = -2$ which can be written as $\frac{-2}{1}$

$\therefore \sqrt[3]{-8}$ is a rational number.

c $-\sqrt{7} = -2.645751311\ldots$ The digits run endlessly without repeating.

$\therefore -\sqrt{7}$ is an irrational number.

d $0.\dot{6} = 0.666\ldots$ which is a recurring decimal

$= \frac{2}{3}$ which is a fraction

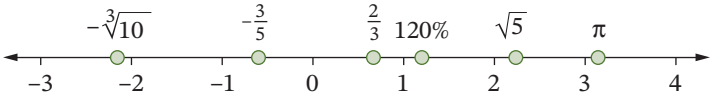
$\therefore 0.\dot{6}$ is a rational number.

e $5\pi = 15.707\ 963\ 27...$ The digits run endlessly without repeating

$\therefore 5\pi$ is an irrational number.

Surds on a number line

The rational and irrational numbers together make up the **real numbers**.
Any real number can be represented by a point on the number line.



$-\sqrt[3]{10} = -2.1544...$ irrational (surd)

$-\frac{3}{5} = -0.6$ rational (fraction)

$\frac{2}{3} = 0.6666...$ rational (fraction)

$120\% = 1.2$ rational (percentage)

$\sqrt{5} = 2.2360...$ irrational (surd)

$\pi = 3.1415...$ irrational (pi)

Example 3

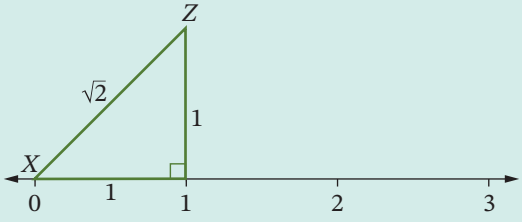
Use a pair of compasses and Pythagoras' theorem to estimate the value of $\sqrt{2}$ on a number line.

Solution

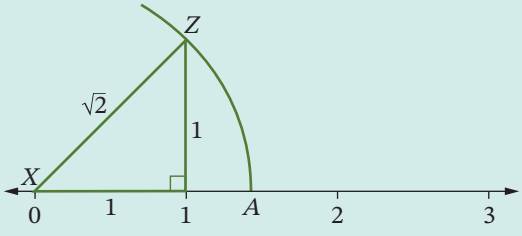
Step 1: Using a scale of 1 unit to 2 cm, draw a number line as shown.



Step 2: Construct a right-angled triangle on the number line with base length and height 1 unit as shown. By Pythagoras' theorem, show that $XZ = \sqrt{2}$ units.



Step 3: With 0 as the centre, use compasses with radius $XZ\left(\sqrt{2}\right)$ to draw an arc to meet the number line at A as shown. The point A represents the value of $\sqrt{2}$ and should be approximately 1.4142...



EXERCISE 13.01 ANSWERS ON P. 637

Surds **U F R C**

1 Which of these numbers is a surd? Select the correct answer **A**, **B**, **C** or **D**. **C**

A $\sqrt{9}$ **B** $\sqrt{225}$ **C** $\sqrt{160}$ **D** $\sqrt{81}$

2 Which of these numbers is NOT a surd? Select **A**, **B**, **C** or **D**. **C**

A $\sqrt{77}$ **B** $\sqrt{144}$ **C** $\sqrt{18}$ **D** $\sqrt{200}$

3 Select the surds from this list of numbers. **C**

$\sqrt{32}$ $\sqrt{33}$ $\sqrt{289}$ $\sqrt{81}$ $\sqrt{4.9}$

$\sqrt{52}$ $\sqrt{121}$ $\sqrt{144}$ $\sqrt{196}$ $\sqrt{200}$

PATH

Worksheet
Surds on the number line

13.01

EXAMPLE 1

Foundation Standard Complex

PATH

EXAMPLE 2

4 Is each number rational (R) or irrational (I)? R C

a $5.\dot{6}$ **b** $\sqrt{8}$ **c** $\sqrt{4}$ **d** $3\frac{1}{7}$

e $\sqrt[3]{27}$ **f** $1.3\dot{5}$ **g** $\sqrt[3]{-64}$ **h** $27\frac{1}{2}\%$

i $\sqrt{5\times10^3}$ **j** $\frac{3}{11}$ **k** $\frac{\sqrt{50}}{3}$ **l** $\sqrt{\sqrt{4}}$

5 Arrange each set of numbers in descending order.

a $1\frac{4}{7}, \sqrt{2}, \frac{\pi}{2}$ **b** $\sqrt[3]{20}, 2.\dot{6}, 2\frac{7}{9}$

6 Express each real number correct to one decimal place and graph them on a number line.

a $-1\frac{4}{5}$ **b** 74% **c** $\frac{4}{11}$ **d** $-\sqrt{12}$

e $-\sqrt[3]{15}$ **f** $2\frac{5}{9}$ **g** $\frac{\pi}{2}$ **h** 187%

7 Use the method from Example **3** to estimate the value of $\sqrt{2}$ on a number line.

EXAMPLE 3

8 **a** Use the method from Example **3** to estimate the value of $\sqrt{5}$ on a number line by constructing a right-angled triangle with base length 2 units and height 1 unit.

b Use a similar method to estimate the following surds on a number line:

i $\sqrt{10}$ **ii** $\sqrt{17}$

13.02 Simplifying surds

PATH

Puzzle Simplifying surds

Technology Simplifying surds quiz

Spreadsheet Simplifying surds quiz

The square of any real number is always positive (except for $0^2 = 0$), so it is not possible to give the square root of a *negative* number.

The radical symbol $\sqrt{\ }$ stands for the *positive* square root of a number. For example $\sqrt{4} = 2$ (not -2).

The square root of x

For $x < 0$ (negative), $\sqrt{x}$ is undefined.

For $x = 0$, $\sqrt{x} = 0$.

For $x > 0$ (positive), $\sqrt{x}$ is the positive square root of x.

For $x \geq 0$, $\left(\sqrt{x}\right)^2 = \sqrt{x} \times \sqrt{x} = x$ and $\sqrt{x^2} = x$.

Foundation Standard Complex

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Example 4

Simplify each expression.

a $\left(\sqrt{7}\right)^2$ **b** $\left(3\sqrt{5}\right)^2$ **c** $\left(-2\sqrt{3}\right)^2$

Solution

a $\left(\sqrt{7}\right)^2 = 7$

b $\left(3\sqrt{5}\right)^2 = 3\sqrt{5} \times 3\sqrt{5}$ $3\sqrt{5}$ means $3 \times \sqrt{5}$

$= 3^2 \times \left(\sqrt{5}\right)^2$

$= 9 \times 5$

$= 45$

c $\left(-2\sqrt{3}\right)^2 = (-2)^2 \times \left(\sqrt{3}\right)^2$

$= 4 \times 3$

$= 12$

The square root of a product

For $x > 0$ and $y > 0$:

$$\sqrt{xy} = \sqrt{x} \times \sqrt{y}$$

A surd $\sqrt{n}$ can be simplified if n can be divided into 2 factors, where one of them is a square number such as 4, 9, 16, 25, 36, 49, ...

Example 5

Simplify each surd.

a $\sqrt{8}$ **b** $\sqrt{108}$ **c** $4\sqrt{45}$ **d** $\dfrac{\sqrt{288}}{3}$

Solution

a $\sqrt{8} = \sqrt{4} \times \sqrt{2}$ 4 is a square number

$= 2 \times \sqrt{2}$

$= 2\sqrt{2}$

b *Method 1*:

$\sqrt{108} = \sqrt{36} \times \sqrt{3}$

$= 6 \times \sqrt{3}$

$= 6\sqrt{3}$

Method 2:

$\sqrt{108} = \sqrt{4} \times \sqrt{27}$

$= 2 \times \sqrt{9} \times \sqrt{3}$

$= 2 \times 3 \times \sqrt{3}$

$= 6\sqrt{3}$

Method 2 involves simplifying surds *twice* ($\sqrt{108}$ and $\sqrt{27}$). Method 1 shows that when simplifying surds, look for the highest square factor possible.

Videos
Simplifying surds 2
Simplifying surds 1

PATH

c $4\sqrt{45} = 4 \times \sqrt{9} \times \sqrt{5}$
$= 4 \times 3 \times \sqrt{5}$
$= 12\sqrt{5}$

d $\frac{\sqrt{288}}{3} = \frac{\sqrt{144} \times \sqrt{2}}{3}$
$= \frac{12\sqrt{2}}{3}$
$= \frac{12^{4}\sqrt{2}}{\cancel{3}^{1}}$
$= 4\sqrt{2}$

EXERCISE 13.02 ANSWERS ON P. 638

Simplifying surds U F R

EXAMPLE 4

1 Simplify each expression.

a $\left(\sqrt{2}\right)^2$ **b** $\left(\sqrt{5}\right)^2$ **c** $\left(3\sqrt{3}\right)^2$ **d** $\left(5\sqrt{10}\right)^2$

e $\left(\sqrt{0.09}\right)^2$ **f** $\left(-2\sqrt{7}\right)^2$ **g** $\left(-3\sqrt{5}\right)^2$ **h** $\left(-5\sqrt{2}\right)^2$

EXAMPLE 5

2 Simplify each surd.

a $\sqrt{50}$ **b** $\sqrt{27}$ **c** $\sqrt{24}$ **d** $\sqrt{54}$

e $\sqrt{243}$ **f** $\sqrt{45}$ **g** $\sqrt{48}$ **h** $\sqrt{200}$

i $\sqrt{96}$ **j** $\sqrt{63}$ **k** $\sqrt{288}$ **l** $\sqrt{108}$

m $\sqrt{75}$ **n** $\sqrt{147}$ **o** $\sqrt{32}$ **p** $\sqrt{242}$

q $\sqrt{162}$ **r** $\sqrt{245}$ **s** $\sqrt{125}$ **t** $\sqrt{512}$

3 Simplify each expression.

a $5\sqrt{50}$ **b** $3\sqrt{8}$ **c** $4\sqrt{27}$ **d** $8\sqrt{98}$

e $\frac{\sqrt{40}}{2}$ **f** $\frac{\sqrt{243}}{9}$ **g** $\frac{\sqrt{28}}{6}$ **h** $3\sqrt{24}$

i $9\sqrt{68}$ **j** $\frac{\sqrt{3125}}{10}$ **k** $\frac{1}{2}\sqrt{72}$ **l** $\frac{3}{4}\sqrt{48}$

m $10\sqrt{160}$ **n** $3\sqrt{75}$ **o** $7\sqrt{68}$ **p** $\frac{\sqrt{52}}{6}$

4 Which surd below is equivalent to $4\sqrt{50}$? Select **A**, **B**, **C** or **D**.

A $8\sqrt{5}$ **B** $20\sqrt{2}$ **C** $8\sqrt{2}$ **D** $20\sqrt{5}$

5 Which surd below is equivalent to $\frac{\sqrt{250}}{10}$? Select **A**, **B**, **C** or **D**.

A $\frac{\sqrt{5}}{10}$ **B** $\frac{\sqrt{10}}{2}$ **C** $2\sqrt{10}$ **D** $5\sqrt{10}$

6 Decide whether each statement is true (T) or false (F). R

a $3\sqrt{7} = \sqrt{21}$ **b** $\sqrt{12} = 6$

c $\left(\sqrt{9.4}\right)^2 = 9.4$ **d** $\sqrt{75} = 5\sqrt{3}$

e $\sqrt{3} \approx 1.7$ **f** The exact value of $\sqrt{10}$ is 3.162 277 8

Foundation Standard Complex

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Adding and subtracting surds

13.03

Just as you can only add or subtract 'like terms' in algebra, you can only add or subtract 'like surds'. You may first need to express all the surds in their simplest forms.

PATH



Puzzle
Surds code puzzle

Example 6

Simplify each expression.

a $4\sqrt{2}+5\sqrt{2}$ **b** $7\sqrt{3}-2\sqrt{3}$ **c** $5\sqrt{2}-3\sqrt{3}+\sqrt{2}$

d $\sqrt{50}+\sqrt{32}$ **e** $\sqrt{8}-\sqrt{27}+\sqrt{18}$ **f** $5\sqrt{20}-3\sqrt{125}$

Solution

a $4\sqrt{2}+5\sqrt{2}=9\sqrt{2}$

b $7\sqrt{3}-2\sqrt{3}=5\sqrt{3}$

c $5\sqrt{2}-3\sqrt{3}+\sqrt{2}=6\sqrt{2}-3\sqrt{3}$

d $\sqrt{50}+\sqrt{32}=\sqrt{25}\sqrt{2}+\sqrt{16}\sqrt{2}$
$=5\sqrt{2}+4\sqrt{2}$
$=9\sqrt{2}$

Simplifying each surd

e $\sqrt{8}-\sqrt{27}+\sqrt{18}=\sqrt{4}\sqrt{2}-\sqrt{9}\sqrt{3}+\sqrt{9}\sqrt{2}$
$=2\sqrt{2}-3\sqrt{3}+3\sqrt{2}$
$=5\sqrt{2}-3\sqrt{3}$

f $5\sqrt{20}-3\sqrt{125}=5\sqrt{4}\sqrt{5}-3\sqrt{25}\sqrt{5}$
$=5\times2\sqrt{5}-3\times5\sqrt{5}$
$=10\sqrt{5}-15\sqrt{5}$
$=-5\sqrt{5}$

PATH

EXERCISE 13.03 ANSWERS ON P. 638

Adding and subtracting surds U F

EXAMPLE 6

1 Simplify each expression.

a $5\sqrt{7}+2\sqrt{7}$ **b** $3\sqrt{2}-8\sqrt{2}$

c $7\sqrt{5}-\sqrt{5}$ **d** $\sqrt{5}+3\sqrt{5}$

e $5\sqrt{17}-5\sqrt{17}$ **f** $3\sqrt{10}-2\sqrt{10}$

g $4\sqrt{15}-3\sqrt{15}+7\sqrt{15}$ **h** $5\sqrt{6}-2\sqrt{6}-4\sqrt{6}$

i $3\sqrt{3}+4\sqrt{3}-5\sqrt{3}$ **j** $4\sqrt{5}+7\sqrt{5}-\sqrt{5}$

k $8\sqrt{10}-5\sqrt{10}+3\sqrt{10}$ **l** $10\sqrt{3}-3\sqrt{3}-12\sqrt{3}$

2 Simplify each expression.

a $3\sqrt{5}-8+2\sqrt{5}$ **b** $11\sqrt{10}+3\sqrt{2}+2\sqrt{10}$

c $-4\sqrt{3}+5\sqrt{2}-5\sqrt{3}$ **d** $3\sqrt{15}+3\sqrt{2}+4\sqrt{15}+5\sqrt{2}$

e $\sqrt{7}-3\sqrt{5}-4\sqrt{7}+\sqrt{5}$ **f** $4\sqrt{6}-3\sqrt{3}-2\sqrt{6}-5\sqrt{3}$

g $10\sqrt{11}-5\sqrt{3}+3\sqrt{11}+4\sqrt{3}$ **h** $\sqrt{13}+8\sqrt{7}-7\sqrt{13}+3\sqrt{7}$

i $2\sqrt{5}-3\sqrt{7}-2\sqrt{5}-3\sqrt{7}$ **j** $4\sqrt{10}-3\sqrt{5}-4\sqrt{10}$

3 For each expression, select the correct simplified answer **A**, **B**, **C** or **D**.

a $\sqrt{3}+\sqrt{12}$

A $5\sqrt{3}$ **B** $\sqrt{15}$ **C** $2\sqrt{6}$ **D** $3\sqrt{3}$

b $4\sqrt{5}-2\sqrt{125}$

A $-6\sqrt{5}$ **B** $\sqrt{5}$ **C** $-\sqrt{45}$ **D** $-46\sqrt{5}$

4 Simplify each expression.

a $\sqrt{8}+\sqrt{32}$ **b** $\sqrt{108}-\sqrt{27}$ **c** $\sqrt{20}-\sqrt{80}$

d $\sqrt{28}-\sqrt{63}$ **e** $3\sqrt{6}+\sqrt{24}$ **f** $2\sqrt{5}+\sqrt{125}$

g $\sqrt{40}-\sqrt{90}$ **h** $5\sqrt{11}+\sqrt{99}$ **i** $3\sqrt{2}+\sqrt{18}$

j $\sqrt{27}+5\sqrt{3}$ **k** $\sqrt{200}-7\sqrt{2}$ **l** $\sqrt{50}+\sqrt{32}$

m $5\sqrt{3}+2\sqrt{27}$ **n** $3\sqrt{20}-\sqrt{245}$ **o** $7\sqrt{12}-5\sqrt{48}$

p $4\sqrt{27}+2\sqrt{243}$ **q** $3\sqrt{63}-2\sqrt{28}$ **r** $2\sqrt{98}+3\sqrt{162}$

s $-5\sqrt{6}+2\sqrt{150}$ **t** $4\sqrt{50}+3\sqrt{18}$ **u** $5\sqrt{27}-6\sqrt{75}$

v $3\sqrt{112}-2\sqrt{252}$ **w** $\sqrt{32}+\sqrt{8}+\sqrt{12}$ **x** $\sqrt{27}+\sqrt{54}+\sqrt{243}$

y $\sqrt{98}-3\sqrt{20}-2\sqrt{8}$ **z** $3\sqrt{96}-2\sqrt{150}+\sqrt{24}$

Foundation Standard Complex

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Multiplying and dividing surds

13.04

The square root of products and quotients

For $x > 0$ and $y > 0$:

$\sqrt{xy} = \sqrt{x} \times \sqrt{y}$

$\sqrt{\frac{x}{y}} = \frac{\sqrt{x}}{\sqrt{y}}$

PATH

13.04

Puzzle Surds

Worksheet Multiplying and dividing surds

Example 7

Simplify each expression.

a $\sqrt{3} \times \sqrt{5}$ **b** $\sqrt{10} \times \sqrt{6}$ **c** $3\sqrt{7} \times 5\sqrt{7}$

d $5\sqrt{27} \times 3\sqrt{6}$ **e** $\sqrt{54} \div (-\sqrt{2})$ **f** $\frac{15\sqrt{32}}{5\sqrt{8}}$

Solution

a $\sqrt{3} \times \sqrt{5} = \sqrt{15}$

b $\sqrt{10} \times \sqrt{6} = \sqrt{60}$
$= \sqrt{4} \times \sqrt{15}$
$= 2\sqrt{15}$

c $3\sqrt{7} \times 5\sqrt{7} = 3 \times 5 \times \sqrt{7} \times \sqrt{7}$
$= 15 \times 7$
$= 105$

d $5\sqrt{27} \times 3\sqrt{6} = 5 \times 3 \times \sqrt{27} \times \sqrt{6}$
$= 15 \times 3\sqrt{3} \times \sqrt{6}$
$= 45\sqrt{18}$
$= 45 \times 3\sqrt{2}$
$= 135\sqrt{2}$

e $\sqrt{54} \div (-\sqrt{2}) = -\frac{\sqrt{54}}{\sqrt{2}}$
$= -\sqrt{27}$
$= -\sqrt{9} \times \sqrt{3}$
$= -3\sqrt{3}$

f $\frac{15\sqrt{32}}{5\sqrt{8}} = 3\sqrt{4}$
$= 3 \times 2$
$= 6$

Example 8

Simplify $\frac{5\sqrt{2} \times 4\sqrt{12}}{10\sqrt{8}}$.

Solution

$\frac{5\sqrt{2} \times 4\sqrt{12}}{10\sqrt{8}} = \frac{20\sqrt{24}}{10\sqrt{8}}$
$= 2\sqrt{3}$

PATH

EXERCISE 13.04 ANSWERS ON P. 638

Multiplying and dividing surds U F

EXAMPLE 7

1 Simplify each expression.

a $\sqrt{7}\times\sqrt{2}$ **b** $-\sqrt{5}\times\sqrt{7}$ **c** $\sqrt{6}\times\sqrt{8}$

d $\sqrt{12}\times\sqrt{3}$ **e** $\sqrt{10}\times\left(-\sqrt{5}\right)$ **f** $3\sqrt{3}\times5\sqrt{3}$

g $5\sqrt{10}\times3\sqrt{3}$ **h** $-2\sqrt{7}\times5\sqrt{3}$ **i** $7\sqrt{5}\times4\sqrt{5}$

j $2\sqrt{3}\times\left(-5\sqrt{6}\right)$ **k** $4\sqrt{3}\times\sqrt{27}$ **l** $-3\sqrt{5}\times4\sqrt{10}$

m $-7\sqrt{2}\times4\sqrt{8}$ **n** $\sqrt{18}\times8\sqrt{3}$ **o** $10\sqrt{2}\times2\sqrt{8}$

p $3\sqrt{18}\times5\sqrt{12}$ **q** $3\sqrt{44}\times\left(-2\sqrt{99}\right)$ **r** $5\sqrt{8}\times4\sqrt{40}$

s $8\sqrt{3}\times3\sqrt{54}$ **t** $-8\sqrt{32}\times\sqrt{27}$ **u** $\sqrt{90}\times\sqrt{72}$

v $-5\sqrt{20}\times3\sqrt{8}$ **w** $7\sqrt{18}\times3\sqrt{24}$ **x** $3\sqrt{48}\times2\sqrt{12}$

2 Simplify each expression.

a $\sqrt{15}\div\sqrt{3}$ **b** $\sqrt{18}\div\left(-\sqrt{6}\right)$ **c** $\dfrac{6\sqrt{48}}{2\sqrt{8}}$

d $10\sqrt{54}\div5\sqrt{27}$ **e** $-3\sqrt{98}\div6\sqrt{14}$ **f** $\dfrac{7\sqrt{18}}{\sqrt{2}}$

g $2\sqrt{24}\div4\sqrt{6}$ **h** $\dfrac{\sqrt{128}}{\sqrt{2}}$ **i** $15\sqrt{18}\div3\sqrt{6}$

j $\dfrac{-20\sqrt{10}}{-4\sqrt{5}}$ **k** $36\sqrt{24}\div9\sqrt{8}$ **l** $16\sqrt{30}\div8\sqrt{5}$

m $12\sqrt{14}\div6$ **n** $\dfrac{3\sqrt{2}}{-12}$ **o** $\sqrt{80}\div4\sqrt{5}$

p $5\sqrt{60}\div\sqrt{15}$ **q** $6\sqrt{8}\div3\sqrt{2}$ **r** $\dfrac{-42\sqrt{54}}{6\sqrt{3}}$

s $12\sqrt{63}\div3\sqrt{7}$ **t** $\dfrac{8\sqrt{50}}{2\sqrt{200}}$ **u** $6\sqrt{3}\div\sqrt{243}$

3 Simplify:

a $\sqrt{6}\times\sqrt{6}$ **b** $\sqrt{7}\times\sqrt{7}$ **c** $2\sqrt{3}\times\sqrt{3}$

d $5\sqrt{y}\times3\sqrt{y}$ **e** $\sqrt{x}\times\sqrt{x}$ **f** $\sqrt{a^2}\times\sqrt{a}$

4 Simplify $3\sqrt{2}\times\sqrt{6}$. Select the correct answer **A**, **B**, **C** or **D**.

A 6 **B** $6\sqrt{2}$ **C** $6\sqrt{3}$ **D** $12\sqrt{2}$

5 Simplify $20\sqrt{10}\div5\sqrt{2}$. Select **A**, **B**, **C** or **D**.

A $4\sqrt{5}$ **B** $15\sqrt{5}$ **C** 10 **D** 20

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6 Simplify each expression.

a $\dfrac{3\sqrt{5}\times 4\sqrt{2}}{3\sqrt{40}}$ **b** $\dfrac{3\sqrt{12}\times 8\sqrt{6}}{4\sqrt{27}}$ **c** $\dfrac{5\sqrt{8}\times 2\sqrt{90}}{10\sqrt{24}}$

d $\dfrac{4\sqrt{5}}{2\sqrt{15}\times 5\sqrt{27}}$ **e** $\dfrac{10\sqrt{686}\times 3\sqrt{12}}{5\sqrt{28}\times\sqrt{18}}$ **f** $\dfrac{8\sqrt{80}\times 3\sqrt{2}}{4\sqrt{5}\times 6\sqrt{8}}$

PATH

EXAMPLE 8

13.05

Binomial products involving surds 13.05

Surd expressions involving brackets can be expanded in the same way as algebraic expressions of the form $a(b + c)$ and $(a + b)(c + d)$.

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Example 9

Expand and simplify each expression.

a $\sqrt{3}\left(\sqrt{5}+\sqrt{7}\right)$ **b** $2\sqrt{11}\left(3\sqrt{11}-5\sqrt{2}\right)$

Solution

a $\sqrt{3}\left(\sqrt{5}+\sqrt{7}\right)=\sqrt{3}\times\sqrt{5}+\sqrt{3}\times\sqrt{7}$

$=\sqrt{15}+\sqrt{21}$

b $2\sqrt{11}\left(3\sqrt{11}-5\sqrt{2}\right)=2\sqrt{11}\times 3\sqrt{11}-2\sqrt{11}\times 5\sqrt{2}$

$=6\times 11-10\times\sqrt{22}$

$=66-10\sqrt{22}$

Example 10

Expand and simplify each binomial product.

a $\left(\sqrt{7}+\sqrt{5}\right)\left(3\sqrt{2}-\sqrt{3}\right)$ **b** $\left(3-2\sqrt{10}\right)\left(\sqrt{5}-3\sqrt{2}\right)$

Solution

a $\left(\sqrt{7}+\sqrt{5}\right)\left(3\sqrt{2}-\sqrt{3}\right)=\sqrt{7}\left(3\sqrt{2}-\sqrt{3}\right)+\sqrt{5}\left(3\sqrt{2}-\sqrt{3}\right)$

$=\sqrt{7}\times 3\sqrt{2}-\sqrt{7}\times\sqrt{3}+\sqrt{5}\times 3\sqrt{2}-\sqrt{5}\times\sqrt{3}$

$=3\sqrt{14}-\sqrt{21}+3\sqrt{10}-\sqrt{15}$

b $\left(3-2\sqrt{10}\right)\left(\sqrt{5}-3\sqrt{2}\right)=3\left(\sqrt{5}-3\sqrt{2}\right)-2\sqrt{10}\left(\sqrt{5}-3\sqrt{2}\right)$

$=3\times\sqrt{5}-3\times 3\sqrt{2}-2\sqrt{10}\times\sqrt{5}+2\sqrt{10}\times 3\sqrt{2}$

$=3\sqrt{5}-9\sqrt{2}-2\sqrt{50}+6\sqrt{20}$

$=3\sqrt{5}-9\sqrt{2}-2\left(5\sqrt{2}\right)+6\left(2\sqrt{5}\right)$

$=3\sqrt{5}-9\sqrt{2}-10\sqrt{2}+12\sqrt{5}$

$=15\sqrt{5}-19\sqrt{2}$

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Special binomial products (see Chapter 12)

Perfect squares

$(a + b)^2 = a^2 + 2ab + b^2$

$(a - b)^2 = a^2 - 2ab + b^2$

Difference of 2 squares

$(a + b)(a - b) = a^2 - b^2$



Video
Binomial products involving surds

Example 11

Expand and simplify each expression.

a $\left(\sqrt{7}-\sqrt{5}\right)^2$

b $\left(2\sqrt{3}+3\sqrt{5}\right)^2$

c $\left(\sqrt{5}-\sqrt{2}\right)\left(\sqrt{5}+\sqrt{2}\right)$

d $\left(3\sqrt{11}+4\right)\left(3\sqrt{11}-4\right)$

Solution

a $\left(\sqrt{7}-\sqrt{5}\right)^2 = \left(\sqrt{7}\right)^2 - 2\times\sqrt{7}\times\sqrt{5} + \left(\sqrt{5}\right)^2$ using $(a - b)^2 = a^2 - 2ab + b^2$

$= 7 - 2\sqrt{35} + 5$

$= 12 - 2\sqrt{35}$

b $\left(2\sqrt{3}+3\sqrt{5}\right)^2 = \left(2\sqrt{3}\right)^2 + 2\times 2\sqrt{3}\times 3\sqrt{5} + \left(3\sqrt{5}\right)^2$ using $(a + b)^2 = a^2 + 2ab + b^2$

$= (4\times 3) + 12\sqrt{15} + (9\times 5)$

$= 12 + 12\sqrt{15} + 45$

$= 57 + 12\sqrt{15}$

c $\left(\sqrt{5}-\sqrt{2}\right)\left(\sqrt{5}+\sqrt{2}\right) = \left(\sqrt{5}\right)^2 - \left(\sqrt{2}\right)^2$ using $(a + b)(a - b) = a^2 - b^2$

$= 5 - 2$

$= 3$

Note that because of the 'difference of 2 squares', the answer is not a surd but a rational number.

d $\left(3\sqrt{11}+4\right)\left(3\sqrt{11}-4\right) = \left(3\sqrt{11}\right)^2 - 4^2$ using $(a + b)(a - b) = a^2 - b^2$

$= (9 \times 11) - 16$

$= 83$

EXERCISE 13.05 ANSWERS ON P. 638

PATH

Binomial products involving surds U F

EXAMPLE 9

1 Expand and simplify each expression.

a $\sqrt{5}\left(\sqrt{3}+\sqrt{2}\right)$ **b** $\sqrt{6}\left(\sqrt{2}-1\right)$ **c** $\sqrt{2}\left(\sqrt{3}+\sqrt{7}\right)$

d $\sqrt{5}\left(3\sqrt{2}-\sqrt{5}\right)$ **e** $3\sqrt{2}\left(\sqrt{2}+2\sqrt{3}\right)$ **f** $-\sqrt{11}\left(4-\sqrt{5}\right)$

g $2\sqrt{7}\left(3\sqrt{7}-4\right)$ **h** $5\sqrt{5}\left(1+3\sqrt{5}\right)$ **i** $3\sqrt{2}\left(4\sqrt{2}+\sqrt{3}\right)$

13.05

EXAMPLE 10

2 Expand and simplify $\left(\sqrt{3}+2\sqrt{5}\right)\left(5\sqrt{2}+\sqrt{3}\right)$. Select the correct answer **A**, **B**, **C** or **D**.

A $20\sqrt{10}$ **B** $2\sqrt{15}+5\sqrt{6}$

C $5\sqrt{6}+3+10\sqrt{10}+2\sqrt{15}$ **D** $5\sqrt{5}+\sqrt{3}+7\sqrt{7}+4\sqrt{2}$

3 Expand and simplify each expression.

a $\left(\sqrt{5}-3\right)\left(2\sqrt{5}+\sqrt{2}\right)$ **b** $\left(\sqrt{7}-\sqrt{3}\right)\left(\sqrt{7}+2\right)$

c $\left(7\sqrt{3}+2\right)\left(4\sqrt{2}+\sqrt{3}\right)$ **d** $\left(3\sqrt{2}-\sqrt{5}\right)\left(5\sqrt{2}+2\sqrt{5}\right)$

e $\left(\sqrt{7}+2\sqrt{11}\right)\left(3\sqrt{7}+4\sqrt{11}\right)$ **f** $\left(5\sqrt{3}-2\sqrt{2}\right)\left(4\sqrt{3}-3\sqrt{2}\right)$

g $\left(6+2\sqrt{10}\right)\left(3\sqrt{10}-1\right)$ **h** $\left(\sqrt{7}-2\sqrt{5}\right)\left(3\sqrt{5}+2\sqrt{7}\right)$

EXAMPLE 11

4 Expand $\left(5+\sqrt{7}\right)^2$. Select **A**, **B**, **C** or **D**.

A 12 **B** 32 **C** $32+10\sqrt{7}$ **D** $32+5\sqrt{7}$

5 Expand and simplify each expression.

a $\left(\sqrt{5}-\sqrt{3}\right)^2$ **b** $\left(\sqrt{7}+\sqrt{2}\right)^2$

c $\left(\sqrt{5}-2\right)^2$ **d** $\left(3+\sqrt{10}\right)^2$

e $\left(5\sqrt{2}+3\sqrt{3}\right)^2$ **f** $\left(5\sqrt{7}-2\right)^2$

g $\left(3\sqrt{2}+2\sqrt{5}\right)^2$ **h** $\left(2\sqrt{5}+\sqrt{3}\right)^2$

6 Expand and simplify each expression.

a $\left(\sqrt{3}-\sqrt{2}\right)\left(\sqrt{3}+\sqrt{2}\right)$ **b** $\left(5+\sqrt{3}\right)\left(5-\sqrt{3}\right)$ **c** $\left(6+2\sqrt{7}\right)\left(6-2\sqrt{7}\right)$

d $\left(\sqrt{5}-\sqrt{3}\right)\left(\sqrt{5}+\sqrt{3}\right)$ **e** $\left(\sqrt{11}-\sqrt{10}\right)\left(\sqrt{11}+\sqrt{10}\right)$ **f** $\left(5\sqrt{7}+3\right)\left(5\sqrt{7}-3\right)$

g $\left(3\sqrt{2}+\sqrt{5}\right)\left(3\sqrt{2}-\sqrt{5}\right)$ **h** $\left(4\sqrt{2}-5\sqrt{3}\right)\left(4\sqrt{2}+5\sqrt{3}\right)$

PATH

7 Expand and simplify $(5\sqrt{2}-4\sqrt{3})(5\sqrt{2}+4\sqrt{3})$. Select **A**, **B**, **C** or **D**.

A $25\sqrt{2}-16\sqrt{3}$ **B** $10\sqrt{2}+10\sqrt{6}$ **C** 2 **D** 26

8 Expand and simplify each expression.

a $(3\sqrt{7}-5)^2$ **b** $(5\sqrt{2}-4)(\sqrt{2}+5)$ **c** $(2\sqrt{7}+3\sqrt{5})(\sqrt{5}+\sqrt{7})$

d $(4\sqrt{3}+5)^2$ **e** $(4\sqrt{2}+\sqrt{3})(4\sqrt{2}-\sqrt{3})$ **f** $(3\sqrt{10}-\sqrt{2})^2$

13.06 Rationalising the denominator

PATH

If $\sqrt{2}\approx 1.4142$, what is the value of $\frac{3}{\sqrt{2}}$? Fractions containing surds in the denominator are difficult to work with. When approximating the value of $\frac{3}{\sqrt{2}}$, it is difficult to mentally divide by 1.4142. We can overcome this by making the denominator **rational** (that is, **not** a surd).

Numbers that have irrational denominators, such as $\frac{1}{\sqrt{5}}, \frac{3}{2\sqrt{7}}, \frac{\sqrt{3}}{\sqrt{2}}, \frac{5\sqrt{7}}{\sqrt{3}}$ can be rewritten with a rational denominator by multiplying both the numerator and denominator by the surd that appears in the denominator. This method is called **rationalising the denominator**.

Video
Rationalising the denominator

Example 12

Rationalise the denominator of each surd.

a $\frac{3}{\sqrt{2}}$ **b** $\frac{5}{4\sqrt{3}}$ **c** $\frac{8\sqrt{2}}{3\sqrt{5}}$ **d** $\frac{\sqrt{2}+1}{\sqrt{3}}$

Solution

a $\frac{3}{\sqrt{2}}=\frac{3}{\sqrt{2}}\times\frac{\sqrt{2}}{\sqrt{2}}$ because $\frac{\sqrt{2}}{\sqrt{2}}=1$

$=\frac{3\sqrt{2}}{2}$

Note: Now it is easier to approximate $\frac{3}{\sqrt{2}}$ by approximating $\frac{3\sqrt{2}}{2}$ and mentally multiplying $\frac{3}{2}$ by 1.4142 than by dividing 3 by 1.4142.

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b $\frac{5}{4\sqrt{3}} = \frac{5}{4\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$

$= \frac{5\sqrt{3}}{4 \times 3}$

$= \frac{5\sqrt{3}}{12}$

c $\frac{8\sqrt{2}}{3\sqrt{5}} = \frac{8\sqrt{2}}{3\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}}$

$= \frac{8\sqrt{10}}{3 \times 5}$

$= \frac{8\sqrt{10}}{15}$

d $\frac{\sqrt{2}+1}{\sqrt{3}} = \frac{\sqrt{2}+1}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$

$= \frac{\sqrt{6}+\sqrt{3}}{3}$

PATH

13.06

EXERCISE 13.06 ANSWERS ON P. 638

Rationalising the denominator U F

EXAMPLE 12

1 By rationalising the denominator, which surd is equivalent to $\frac{2}{\sqrt{6}}$?
Select the correct answer **A**, **B**, **C** or **D**.

A $2\sqrt{6}$ **B** $\frac{\sqrt{6}}{3}$ **C** $\frac{\sqrt{6}}{6}$ **D** $\frac{2\sqrt{6}}{3}$

2 Rationalise the denominator of each surd.

a $\frac{1}{\sqrt{2}}$ **b** $\frac{1}{\sqrt{7}}$ **c** $\frac{1}{\sqrt{3}}$ **d** $\frac{3}{\sqrt{2}}$

e $\frac{2}{\sqrt{7}}$ **f** $\frac{1}{3\sqrt{2}}$ **g** $\frac{1}{2\sqrt{3}}$ **h** $\frac{1}{4\sqrt{7}}$

i $\frac{7}{3\sqrt{5}}$ **j** $\frac{\sqrt{2}}{3\sqrt{5}}$ **k** $\frac{3\sqrt{2}}{2\sqrt{6}}$ **l** $\frac{5\sqrt{3}}{4\sqrt{5}}$

3 Which surd is equivalent to $\frac{\sqrt{3}}{2\sqrt{5}}$? Select **A**, **B**, **C** or **D**.

A $\frac{\sqrt{15}}{10}$ **B** $2\sqrt{15}$ **C** $\frac{\sqrt{15}}{3}$ **D** $\sqrt{5}$

4 Which surd is equivalent to $\frac{\sqrt{27}}{3\sqrt{18}}$? Select **A**, **B**, **C** or **D**.

A $\frac{1}{2}$ **B** $\frac{\sqrt{2}}{2}$ **C** $\frac{\sqrt{5}}{6}$ **D** $\frac{\sqrt{6}}{6}$

5 Rationalise the denominator of each expression.

a $\frac{\sqrt{2}-1}{\sqrt{2}}$ **b** $\frac{1-\sqrt{5}}{\sqrt{5}}$ **c** $\frac{5+\sqrt{3}}{2\sqrt{2}}$ **d** $\frac{\sqrt{2}-\sqrt{3}}{3\sqrt{6}}$

6 Simplify each expression, giving the answer with a rational denominator.

a $\frac{1}{\sqrt{7}} + \frac{1}{\sqrt{2}}$ **b** $\frac{\sqrt{2}}{\sqrt{5}} + \frac{3}{\sqrt{3}}$ **c** $\frac{3}{2\sqrt{3}} - \frac{1}{\sqrt{2}}$

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